MRinRT 2026

Status, Challenges and Opportunities

PROGRAMME

Wednesday 17.6.2026

8.00-8.45: Opening and Welcome

8.45-9.30: Keynote: *The Clinical Evidence for MRI in Radiation Therapy*
Uulke van der Heide, Netherlands Cancer Institute, Amsterdam, The Netherlands

Break (30 min)

State of the art solutions for MRI-guided RT

10.00-10.25: *Offline MRI-guided RT*
Neelam Tyagi, Memorial Sloan Kettering Cancer Center, New York, NY, USA

10.25-10.50: *Online MRI-guided RT*
Sebastian Klüter, Heidelberg University Hospital, Heidelberg, Germany

10.50-11.30: Proffered papers 1

27	Kiaran McGee	Teaching an old dog new tricks: developing customized RF coils for radiation therapy MR simulation — considerations and experiences.
36	Mizgin Coskun	Metal artifact reduction for 3T MRI-only prostate radiotherapy with hip prostheses: target definition, synthetic CT and fiducial marker identification
91	Søren Agergaard	Treating intermediate-risk prostate cancer with no prior patient-specific imaging using online adaptive MRI-guided radiotherapy
87	Hubert Gabrys	Automation enables efficient MRI-enhanced online adaptive RT on a conventional C-arm linac

Break (15 min)

Technical and physics innovations

11.45-12.45: Proffered papers 2

70	Kathryn Mittauer	Plan quality reporting for single-fraction, stereotactic MR-guided adaptive radiotherapy
6	Thomas Willigenburg	Treatment efficiency and dosimetric benefit of intrafraction motion management during MRI-guided SBRT for liver tumors
69	Matteo Nardini	MRI Toolkit for 0.35 T MR-Linac Adaptive Workflows: Toward Robust Quantitative Biomarkers for dose painting
100	Nikolai Mickevicius	Target-driven acquisition and super-resolution reconstruction for simultaneous cine imaging and 4D-MRI

95	Nicolas Cote	Deformable Dose Accumulation Variability Analysis in Localized Pancreatic Cancer Treated with MR-Guided Adaptive Radiotherapy
68	Robert Adam Mitchell	Motion management application in two-fraction prostate radiotherapy: initial DESTINATION 2 trial experience

12.45-13.45: Lunch

Clinical trials 1

13.45-14.20: *Clinical Trials - Why they are important, what to consider and how to design*
Søren Bentzen, University of Maryland School of Medicine Baltimore, Maryland, USA

14.20-15.00: Proffered papers 3

49	Valentina Lancellota	MR-image-guided adaptive interventional radiotherapy boost for vaginal recurrences
11	Pia Hartfelt	Adaptive MRI-guided Stereotactic Body Radiation Therapy for pancreatic cancer: Clinical outcomes in a prospective cohort
83	Sabine Linden	MR-guided ultra-hypofractionated radiotherapy with real-time motion management for early-stage glottic cancer patients
65	Angela Romano	Clinical Outcomes of MRI-guided Pelvic Reirradiation in Previously Irradiated Locally Recurrent Rectal Cancer

Break (15 min)

AI-based workflow improvements

15.15-15.40: *AI for radiation therapy*
Lorenzo Placidi, Fondazione Policlinico Universitario A. Gemelli IRCCS, Rome, Italy

15.40-16.30: Proffered papers 4

97	Nina Tilly	From motion to dose: Dosimetric evaluation of motion-based margins using AI-predicted dose in MR-Linac adaptive workflows
5	Musti Kadhim	Catching MRI outliers: unsupervised detection and localization of MRI artefacts and clinical anomalies using deep learning
17	Nikolaos Delopoulos	Eliminating synthetic CT? Fast AI dose calculation directly on MRI
23	Elisabeth Kildegaard	AI-based organ segmentation to streamline MRI-guided adaptive radiotherapy for locally advanced pancreatic cancer
103	Anil Sethi	Predictors of Treatment Plan Adaptation in Radiation Therapy for Pancreatic Cancer

Break (15 min)

Young investigator award presentations

16.45-18.00: Proffered papers 5

44	Chenyang LIU	AI-empowered Four-dimensional Magnetic Resonance Fingerprinting (AIE-4DMRF) for Accurate and Efficient Liver Cancer Radiotherapy Motion Management
86	Ryan Bonate	Repeatability coefficient thresholding for voxelwise qMRI treatment response in head and neck cancer
24	Jamila Guichelaar	Microstructural Assessment of Oral Squamous Cell Carcinoma Using Time-Dependent Diffusion MRI
13	Minoo Gandomi	Diffusion-MRI hypoxia mapping with distortion correction in head and neck cancer radiotherapy
7	Yimin Ni	Motion-Corrected Multimodality MRF for Pseudo-CT Generation: A Feasibility Study in Abdominal Radiotherapy
15	Bloem van Dam	Low cost, high reward: real-time motion management for pancreatic tumors on the 1.5 T MR-Linac
98	Mehdi Shojaei	Multi-contrast conditional organs-at-risk segmentation for online adaptive MR-guided radiotherapy of pancreatic cancer

18.30- : Social event and symposium reception (drinks and finger food)

Thursday 18.6.2026

MRI for Biological Guidance in Radiation Therapy

8.30-8.55: Tentative: *The current evidence and future directions for MRI biomarkers in radiation therapy*
Lars Olsson, Lars Olsson, Lund University , Lund, Sweden

8.55-9.35: Proffered papers 6

4	Ayesha Das	Mathematical modeling of MRI-based tumor habitats during radiotherapy in a murine model of glioma
54	Erik van der Bijl	Multi-institutional in-vivo characterization of consensus measurement methods for T1 and T2 mapping on the 1.5T Unity MR-linac
30	Carlo Guglielmo Cattaneo	Evaluation of the Combined Effects of Magnetic Field and Radiotherapy on Patient-Derived Pancreatic Ductal Adenocarcinoma Organoids
66	Matteo Nardini	Longitudinal ADC changes on a 0.35 T MR-Linac during rectal cancer MRI-guided radiotherapy

Break (15 min)

Quantitative MRI 1: Clinical and Physics Applications

9.50-10.15: TBD

10.15-11.05: Proffered papers 7

25	Anne Bisgaard	Prognostic value of diffusion-weighted MRI and clinical parameters in prediction of overall survival after SBRT in patients with pancreatic cancer
14	Edoardo Salmeri	Spatiotemporal Deep Learning for Early Prediction of Glioblastoma Response Using 1.5T MR-Linac Imaging
20	Walid Dandachly	Multiparametric MRI–Based Habitat Imaging for Risk Stratification and Survival Prediction in Head and Neck Tumors
39	Lena Kretzschmar	Prospective study on response assessment during MR-guided RT for glioblastoma – first results from the MARGA-I-study
79	Sirisha Tadimalla	Development and validation of MRI biomarkers of liver function radiation response for functional-guided liver SBRT

Break (15 min)

Advancements in motion management

11.20-11.45: *Basics of MRI and motion management*
Jie Deng, UT Southwestern Medical Center, Dallas, USA

11.45-12.45: Proffered papers 8

88	Eric Paulson	Toward augmented reality-driven, MR-guided interstitial brachytherapy with active tracking
72	Yuqing Xiong	4D motion propagation from orthogonal cine MRI for time-resolved volumetric imaging in MR-guided radiotherapy
67	Frédérique van Gameren	GUIDING-U trial: Effect of audio-visual breathing guidance on improving scan efficiency for respiratory-correlated abdominal and thoracic MRI
90	Nikolai Mickevicius	Comparison of pilot tone- and self-navigated stack-of-stars 4D-MRI
101	Eric Paulson	Feasibility of ADC-based intrafraction biological target tracking for MR-guided radiotherapy
52	Lieke Meijers	More Contrast, Less artifacts: Redefining cine MRI for abdominal and esophageal treatments in the MR-Linac workflow

12.45-13.45: Lunch

Clinical trials 2

13.45-14.25: Proffered papers 9

51	Michael Chuong	Long-term efficacy and safety of ablative single-fraction stereotactic MR-guided adaptive radiotherapy from the multi-center phase 2 SMART ONE trial
53	Angela Romano	Magnetic resonance response–driven adaptive radiotherapy for locally advanced rectal cancer: long-term results of the THUNDER-2 trial (NCT04815694)

78	Kristine Skovly Nielsen	MR-Linac–based treatment of oligorecurrent prostate cancer: patient characteristics and treatment patterns from the international MOMENTUM study
50	Valentina Lancellotta	Radio-chemotherapy and MR-Image Guided Interventional Radiotherapy in vaginal carcinoma: a monocentric experience

Break (5 min leg stretch)

Multi disciplinarity and national collaborations

14.30-14.55: Tentative: *Leveraging institutional MRI expertise for radiotherapy*
Trina Herbert, The Royal Marsden NHS Foundation Trust, London, UK

14.55-15.20: TBD, *National safety initiatives*

15.20-15.35: TBD, *National MRI QA network (Nimbus)*

15.35-15.45: Mette Felter, *A Virtual MR-linac Center: Equal Access, Shared Innovation and Expertise*

Panel discussion

Break (15 min)

Contrast agents and novel tissue contrast

16.15-16.45: Tentative: *MRI contrast mechanism, safety and novel agents*
Jan Henrik Ardenkjær, Technical University of Denmark, Kgs Lyngby, Denmark

Vendor updates

16.45-16.55: Elekta

16.55-17.05: Siemens

17.05-17.15: Bracco

Break (15 min)

Poster pitches

17.30-18.30: Proffered papers 10 (2 min pitch)

59	Elisabeth Kildegaard	Maintaining Image Quality with Less Gadolinium: Implementation of Vueway in Radiotherapy MRI Protocols
58	Giulia Panza	REPAIR-Lung: REPeAted mrI-guided stereotactic body Radiotherapy (reSBRT) for lung oligometastases, a mono-institutional retrospective analysis
34	Erin Wang	Hypoxia imaging in prostate cancer with innovative methods: the HYPOXIM clinical trial

35	Carlo Guglielmo Cattaneo	Early ctDNA dynamics impact in locally advanced rectal cancer: interim analysis of the MOREOVER study
77	Erik van der Bijl	MRI guided adaptive stereotactic radiotherapy for endometrical cancer: feasibility
99	Eric Paulson	Feasibility of online 4D-MRI within five minutes for MR-guided adaptive radiotherapy
89	Josiah Simeth	Distance-to-contour conditioned coregistration for Pancreatic cancer RT
42	Peilin Wang	Few-Shot Adaptation for Multi-Contrast Four-Dimensional Motion Reconstruction in Abdominal MR-guided Radiotherapy
12	Uffe Bernchou	Clinical robustness of online adaptive 1.5T MRI-guided pancreatic SBRT: Baseline vs adaptive plan quality
47	Christopher Kurz	Accounting for cropped training data for MR-linac autosegmentation using the Tversky loss
48	Valentina Lancellotta	Enomogram to predict the rate of loco-regional control after radiochemotherapy and interventional radiotherapy in cervical cancer
92	Christopher Kehayias	Non-breathing Abdominal Organ-At-Risk Motion Dynamics Characterized Using Golden-Angle Radial Samples and Hierarchical Motion Modeling
56	Yulia Shcherbakova	Optimization of small bowel cine-imaging for fast 3D tracking during MR-guided Radiotherapy
62	Jan Saip Aunan-Diop	Magnetic resonance elastography enables lesion-level separation of radiation necrosis and recurrent brain metastasis
74	Peter Latimer	Investigating the apparent diffusion coefficient as a biomarker of treatment response in pancreas tumours on an MR-Linac
29	Ancuta Friismose	Longitudinal magnetic resonance elastography reveals post-treatment viscoelastic changes in glioblastoma
85	Chris Moore	Oxygen-enhanced TOLD and BOLD MRI in prostate cancer and arterial and venous blood.
84	David Tilly	Comparing local and external nnU-Net segmentation models for online prostate MRgRT
55	Ivan A. Rashid	Longitudinal Q-space trajectory imaging in patients with prostate cancer receiving definitive radiotherapy

18.30-20.00: Poster viewing and tapas

Friday 19.6.2026

Quality Assurance and Image Quality

8.30-9.00: Tentative: *MRI QA recommendations and practical implementation*
Joan Chick, University College London Hospitals NHS Foundation Trust, London, UK

9.00-9.50: Proffered papers 11

96	Stefan Dorsch	Towards MR-guided particle therapy: Evaluation of polymer gel for geometrical validations with online readout
71	Florian Putz	An Image-Based Method to Detect Missing Distortion Correction for Gradient Non-Linearities in Externally Acquired Planning MRIs

81	Bernd-Niklas Axer	Suitability of human body donor heads for quality assurance in MR-only radiotherapy
28	Anders Bertelsen	Optimizing enema and bladder filling procedures for ultra-hypofractionated online MR guided adaptive prostate radiotherapy
22	Frédérique van Gameren	Impact of MRI triggering strategies on exhale-position image accuracy under realistic respiratory motion conditions

Break (15 min)

Quantitative MRI 2: Optimization and Low-field Solutions

10.05-10.30: Tentative: *How to get the most out of low field MRI systems*
Najat Salameh, University of Aberdeen, Aberdeen, UK

10.30-11.30: Proffered papers 12

18	Julian van Oorschodt	Accuracy of accelerated T1 and T2 relaxometry sequences and repeatability in pelvic tissues
41	Xiang Wang	Acquisition optimization for reliable quantitative MRF in MR-guided radiotherapy
75	Damien McHugh	Feasibility of high b-value diffusion-weighted imaging on an MRI-linac to improve prostate lesion conspicuity
26	Hassan Bagher-Ebadian	Overcoming Low-Field Limitations in Quantitative Brain DCE-MRI Using Protocol Optimization and Probabilistic Model Selection
76	Prashant Prabhakaran Nair	Technical Validation of Accelerated Diffusion-weighted MRI on a 1.5T MR-Linac
57	Magali Nuixe	Correcting slice profile and B1 imperfections to improve the accuracy of MR fingerprinting T1 and T2 mapping on a 1.5 T MR-Linac

11.35-12.00: Host for MRinRT 2028

12.00-12.30: Awards and closing remarks

Lunch boxes and adjourn